

Project Report: Designing and Implementing a Simplified Data Warehouse Using open library book data

Dataset Overview

The Open Library dataset contains several files that offer detailed metadata related to books, their ratings, authors, and editions. This report examines the following specific files: **works**, **ratings**, **editions**, and **authors**.

Dataset Structure

- **File Format:** JSON
 - **Primary Files:**
 - **Works:** This file holds information about books, such as titles, authors, subjects, and additional metadata.
 - **Ratings:** This file provides ratings for books, with a link to the corresponding work via the `work_id`.
 - **Editions:** This file includes data on various editions of works, with details about each edition.
 - **Authors:** This file contains data on authors, including their names and the works they are linked to.
-

Key Fields

1. **Works Dataset:**
 - `work_id`: Unique identifier for each work.
 - `title`: Title of the work.
 - `author_id`: List of author IDs associated with the work (foreign key referencing `dim_authors`).
 - `revision`: Topics or genres of the work updates.
 - `covers`: Topics or genres of the work.
2. **Ratings Dataset:**
 - `work_id`: Links the rating to a specific work (foreign key referencing `dim_works`).
 - `rating`: Numerical rating assigned to the work.
 - `rating_count`: The number of ratings the work has received.
 - `date`: Date when the rating was given.
3. **Editions Dataset:**
 - `book_id`: Unique identifier for each edition (primary key).

- **work_id:** Links the edition to a work (foreign key referencing `dim_works`).
 - **title:** Title of the edition.
 - **authors:** Authors associated with this specific edition.
 - **publish_date:** Publication date of the edition.
 - **isbn_13:** ISBN-13 of the book edition.
 - **physical_format:** Format of the book edition (e.g., hardcover, paperback).
4. **Authors Dataset:**
- **author_id:** Unique identifier for each author (primary key).
 - **name:** Name of the author.
 - **revision:** The revision number of the author's data.
 - **created_at:** Timestamp when the author data was first created.
 - **key:** A unique key for the author (likely from an external system like Open Library).
 - **source_records:** Source records associated with the author (can be a list of source identifiers).
 - **latest_revision:** Latest revision number for the author's data.
 - **last_modified:** Timestamp of when the author's data was last modified.
-

Dataset Content

- **Works Dataset:** Contains comprehensive metadata about books in the Open Library catalog, with key fields for book titles, subjects, and author information.
 - **Ratings Dataset:** Contains ratings for works; however, many works may lack ratings due to insufficient data or a lack of ratings altogether.
 - **Editions Dataset:** Includes various editions of books, providing more detailed data about specific versions of a work.
 - **Authors Dataset:** Contains author information, linking them to the works they have created or contributed to.
-

Initial Observations

1. **Missing Data:**
 - The **works dataset** may have missing fields for subjects or authors, and some works may not have a `published_date`.
 - The **ratings dataset** may lack rating values for certain works, potentially due to insufficient data or the absence of ratings.
 - The **editions dataset** may have missing or inconsistent data, particularly with respect to authors or publication dates.
 - The **authors dataset** may be incomplete, as some authors may not have associated works listed.
2. **Data Types:**

- There are inconsistent formats, especially with date fields such as `published_date`, which are stored as strings and will need to be converted into proper date types.
 - In the **ratings dataset**, both `rating` and `rating_count` are numeric but may be stored as strings, requiring conversion.
 - 3. **Nested Structures:**
 - The `authors` field in the **works** and **editions** datasets is a list of IDs, which will need to be linked to the **authors** dataset for proper association.
 - The `subjects` field in the **works** dataset is a list of strings, requiring transformation into a more normalized structure for efficient querying.
 - 4. **Duplicates:**
 - Preliminary checks suggest no obvious duplicate rows in the datasets. However, duplicates across tables (e.g., multiple editions of the same work) may require handling during the ETL (Extract, Transform, Load) process.
 - 5. **Data Quality:**
 - Certain fields are either missing or have inconsistent formats (e.g., `published_date` and `rating`). These fields will need to be cleaned and standardized during the ETL process.
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Dimensional Modeling

To design a simplified star schema for the Open Library dataset, the focus will be on the fact table and the associated dimension tables for **authors**, **works**, **ratings**, and **editions**.

1. Star Schema Design

The star schema will consist of a central fact table (`fact_book_rating`) surrounded by dimension tables. The structure of the schema includes:

Fact Table

- **fact_book_ratings:** Contains aggregated data about books, with references to associated works, editions, ratings, and authors.
 - `book_info_id` (Primary Key): Unique identifier for each book record.
 - `book_id` (Foreign Key to `dim_editions`): Refers to the book edition associated with the rating.
 - `Title` :name of the title of the book
 - `work_id` (Foreign Key to `dim_works`): Refers to the work associated with the book.
 - `author_id` (Foreign Key to `dim_authors`): Refers to the author(s) associated with the book.
 - `author_name`: Name of the author associated with the book.
 - `rating`: Numeric score representing the rating given to the book.

Dimension Tables

- **dim_editions:** Stores information about book editions.
 - **book_id** (Primary Key): Unique identifier for each book edition.
 - **title:** Title of the book edition.
 - **isbn_13:** ISBN-13 of the book edition.
 - **isbn_10:** ISBN-10 of the book edition.
 - **publishers:** Publisher of the book edition.
 - **publish_date:** Date when the book was published.
 - **physical_format:** Format of the book edition (e.g., hardcover, paperback).
 - **number_of_pages:** Total number of pages in the book edition.
 - **authors:** Authors of the book edition.
 - **type:** Type of edition (e.g., first edition, revised edition).
 - **revision:** Revision number of the book edition.
 - **last_modified:** Timestamp of the last modification.
- **dim_works:** Stores information about literary works.
 - **work_id** (Primary Key): Unique identifier for each work.
 - **title:** Title of the work.
 - **author_id** (Foreign Key to dim_authors): Identifier(s) of the author(s) of the work.
 - **subject:** Subject(s) of the work (may include multiple subjects).
 - **created_at:** Date when the work was created.
- **dim_ratings:** Stores ratings information for books.
 - **rating_id** (Primary Key): Unique identifier for each rating.
 - **book_id** (Foreign Key to dim_editions): Identifier for the book being rated.
 - **work_id** (Foreign Key to dim_works): Identifier for the work associated with the rating.
 - **rating:** Rating score given to the book.
 - **date:** Date when the rating was given.
- **dim_authors:** Stores information about authors.
 - **author_id** (Primary Key): Unique identifier for each author.
 - **name:** Name of the author.
 - **revision:** The revision number of the author's data.
 - **created_at:** Timestamp when the author data was first created.
 - **key:** A unique key for the author (likely from an external system like Open Library).
 - **source_records:** Source records associated with the author (can be a list of source identifiers).
 - **latest_revision:** Latest revision number for the author's data.
 - **last_modified:** Timestamp of when the author's data was last modified.

2) Table Schema Definitions

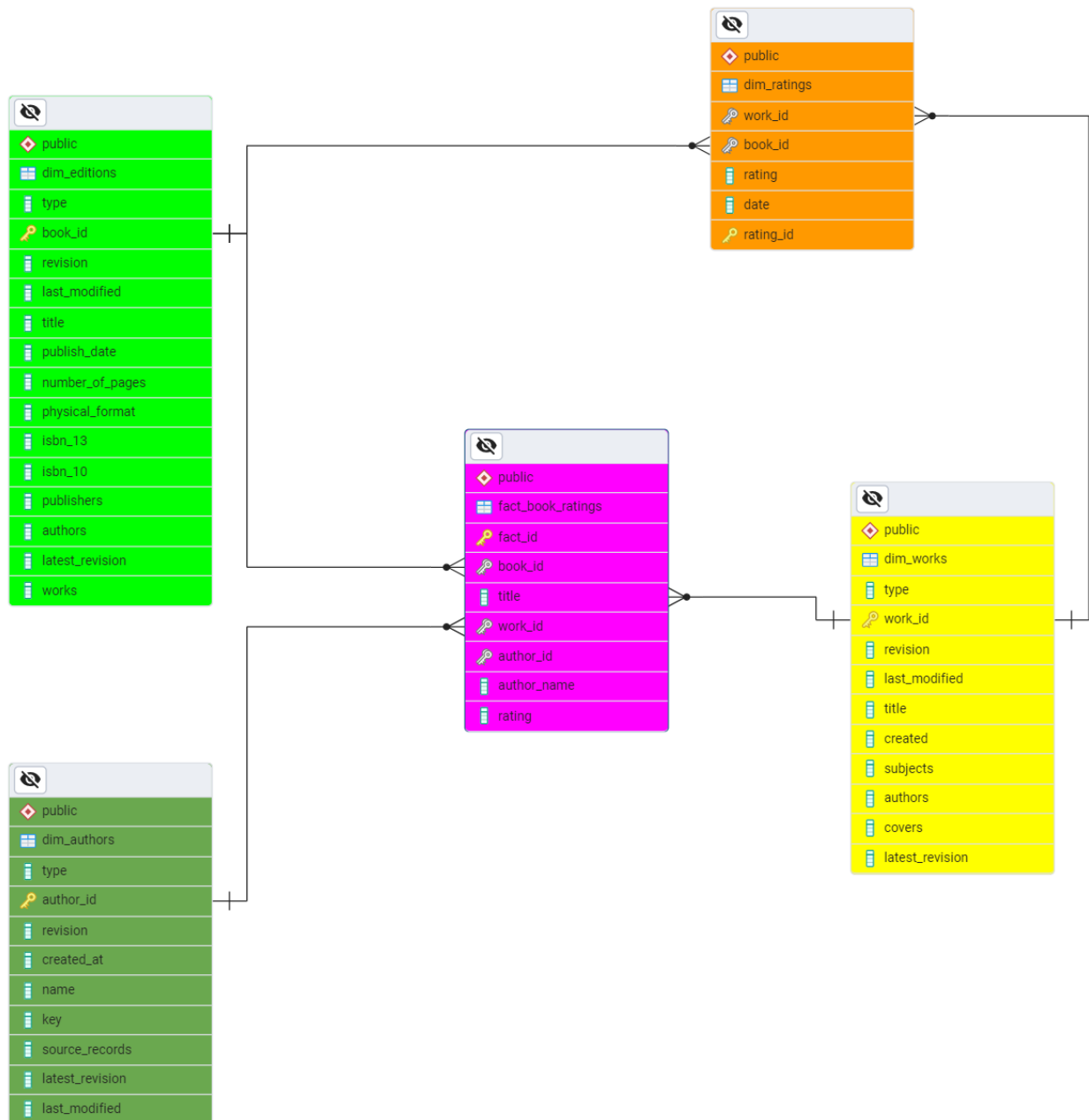
```
CREATE TABLE fact_book_ratings
(
    book_id VARCHAR,
    title VARCHAR(255),
    work_id VARCHAR,
    author_id VARCHAR,
    author_name VARCHAR(255),
    rating NUMERIC,
    CONSTRAINT fact_book_ratings_author_id_fkey FOREIGN KEY (author_id)
        REFERENCES dim_authors (author_id) MATCH SIMPLE
        ON UPDATE CASCADE
        ON DELETE CASCADE,
    CONSTRAINT fact_book_ratings_book_id_fkey FOREIGN KEY (book_id)
        REFERENCES dim_editions (book_id) MATCH SIMPLE
        ON UPDATE CASCADE
        ON DELETE CASCADE,
    CONSTRAINT fact_book_ratings_work_id_fkey FOREIGN KEY (work_id)
        REFERENCES dim_works (work_id) MATCH SIMPLE
        ON UPDATE CASCADE
        ON DELETE CASCADE
);
CREATE TABLE dim_editions
(
    type VARCHAR(255),,
    book_id VARCHAR(255),NOT NULL,
    revision BIGINT,
    last_modified TIMESTAMP WITHOUT TIME ZONE,
    title VARCHAR(255),,
    publish_date TIMESTAMP WITHOUT TIME ZONE,
    number_of_pages DOUBLE PRECISION,
    physical_format VARCHAR,
    isbn_13 VARCHAR(255),,
    isbn_10 VARCHAR(255),,
    publishers VARCHAR,
    authors VARCHAR(255),,
    latest_revision DOUBLE PRECISION,
    works VARCHAR(255),,
    CONSTRAINT dim_editions_pkey PRIMARY KEY (book_id)
);
CREATE TABLE dim_ratings
(
    work_id VARCHAR,
    book_id VARCHAR,
    rating BIGINT,
    date TIMESTAMP WITHOUT TIME ZONE,
    rating_id BIGINT NOT NULL DEFAULT
nextval('dim_ratings_rating_id_seq'::regclass),
    CONSTRAINT dim_ratings_pkey PRIMARY KEY (rating_id),
    CONSTRAINT dim_ratings_work_id_fkey FOREIGN KEY (work_id)
        REFERENCES dim_works (work_id) MATCH SIMPLE
        ON UPDATE CASCADE
        ON DELETE CASCADE
    NOT VALID,
    CONSTRAINT g FOREIGN KEY (book_id)
```

```

        REFERENCES dim_editions (book_id) MATCH SIMPLE
        ON UPDATE CASCADE
        ON DELETE CASCADE
        NOT VALID
);
CREATE TABLE dim_works
(
    type VARCHAR,
    work_id VARCHAR NOT NULL,
    revision BIGINT,
    last_modified TIMESTAMP WITHOUT TIME ZONE,
    title VARCHAR,
    created TIMESTAMP WITHOUT TIME ZONE,
    subjects VARCHAR,
    authors VARCHAR,
    covers VARCHAR,
    latest_revision BIGINT,
    CONSTRAINT dim_works_pkey PRIMARY KEY (work_id)
);
CREATE TABLE dim_authors
(
    type VARCHAR,
    author_id VARCHAR NOT NULL,
    revision BIGINT,
    created_at TIMESTAMP WITHOUT TIME ZONE,
    name VARCHAR,
    key VARCHAR,
    source_records VARCHAR,
    latest_revision DOUBLE PRECISION,
    last_modified TIMESTAMP WITHOUT TIME ZONE,
    CONSTRAINT dim_authors_pkey PRIMARY KEY (author_id)
);

```

3 STAR schema



3. ETL Design:

```
1 import pandas as pd
2 from sqlalchemy import create_engine
3 from sqlalchemy.exc import SQLAlchemyError
4 import os
5 import json
6 import re
7
8 # Specify the path to your file
9 FILE_PATH = r'D:\datasets\vol_dump_editions_2025-03-31.txt'
10
11 # PostgreSQL database connection parameters
12 DB_PARAMS = {
13     'dbname': 'Second DB',
14     'user': 'postgres',
15     'password': '1236', # Ensure this password is correct
16     'host': 'localhost',
17     'port': '5432'
18 }
19
20
21 def extract_data(file_path):
22     """Extract data from the editions dump file."""
23     data = []
24     try:
25         with open(file_path, 'r', encoding='utf-8') as file:
```

2

```
19
20
21 def extract_data(file_path):
22     """Extract data from the editions dump file."""
23     data = []
24     try:
25         with open(file_path, 'r', encoding='utf-8') as file:
26             for line in file:
27                 parts = line.strip().split('\t')
28                 if len(parts) >= 5: # Ensure there are enough columns
29                     json_data = json.loads(parts[4]) # Parse the JSON part
30                     data.append({
31                         'type': parts[0],
32                         'book_id': parts[1],
33                         'revision': int(parts[2]),
34                         'last_modified': parts[3],
35                         'title': json_data.get('title'),
36                         'publish_date': json_data.get('publish_date'),
37                         'number_of_pages': json_data.get('number_of_pages'),
38                         'physical_format': json_data.get('physical_format'),
39                         'isbn_13': json_data.get('isbn_13'),
40                         'isbn_10': json_data.get('isbn_10'),
41                         'publishers': json_data.get('publishers'),
42                         'authors': json_data.get('authors'),
43                         'latest_revision': json_data.get('latest_revision').
```

3

```
59 def clean_data(data):
60     """Clean the extracted data."""
61     try:
62         # Clean the data by removing extra characters
63         data['type'] = data['type'].replace('/type/', '', regex=True)
64         data['book_id'] = data['book_id'].replace('/books/', '', regex=True)
65
66         # Clean JSON-like fields to remove extra quotes and commas
67         data['isbn_13'] = data['isbn_13'].apply(
68             lambda x: ', '.join(isbn.strip('\"') for isbn in x) if isinstance(x, list)
69         )
70         data['isbn_10'] = data['isbn_10'].apply(
71             lambda x: ', '.join(isbn.strip('\"') for isbn in x) if isinstance(x, list)
72         )
73         data['publishers'] = data['publishers'].apply(
74             lambda x: ', '.join(publisher.strip('\"') for publisher in x) if isinstance(x, list)
75         )
76         data['authors'] = data['authors'].apply(
77             lambda x: ', '.join(author.get('key').replace('/authors/', '') for author in x) if isinstance(x, list)
78         )
79
80         data['works'] = data['works'].apply(
81             lambda x: ', '.join(work.get('key').replace('/works/', '') for work in x) if isinstance(x, list)
82         )
83
84         # Clean date formats
85         data['last_modified'] = pd.to_datetime(data['last_modified'], errors='coerce')
86         data['publish_date'] = pd.to_datetime(data['publish_date'], errors='coerce')
87
88         # Remove duplicates based on 'book_id'
89         data = data.drop_duplicates(subset=['book_id'])
```

4

```
17 def transform_data(data):
18     return pd.DataFrame()
19
20
21 def load_data(data, db_params):
22     """Load data into PostgreSQL database."""
23     try:
24         engine = create_engine(f'postgresql://{db_params["user"]}:{db_params["password"]}@{db_params["host"]}:{db_params["port"]}/{db_params["dbname"]}')
25
26         data.to_sql('dim_editions', con=engine, if_exists='replace', index=False, chunksize=500)
27         print("Data loaded successfully into the database!")
28     except SQLAlchemyError as e:
29         print("Database error: (e)")
30     except Exception as e:
31         print("An error occurred while loading data: (e)")
32
33
34 def main():
35     """Main function to run the ETL process."""
36     if not os.path.exists(FILE_PATH):
37         print(f"Error: The specified file (FILE_PATH) was not found.")
38     else:
```

```
def extract_data(file_path):
    'last_modified': json_data.get('last_modified', {}).get('value')
    })
    if len(data) >= 500000: # Limit to 500,000 rows
        break

Run authors EETL.py

C:\Users\Hailsha\PycharmProjects\pythonProject\venv\Scripts\python.exe "C:\Users\Hailsha\PycharmProjects\Data Mining\authors EETL.py"
type ... last_modified
0 /type/author ... 2021-12-26T21:42:09.238563
1 /type/author ... 2021-12-26T22:24:12.384187
2 /type/author ... 2021-12-26T23:32:38.430686
3 /type/author ... 2021-12-27T00:25:38.211432
4 /type/author ... 2021-12-27T00:25:45.312956
...
499995 /type/author ... 2022-11-22T00:24:07.209908
499996 /type/author ... 2022-11-22T00:25:32.944356
499997 /type/author ... 2022-11-22T00:31:34.553625
499998 /type/author ... 2022-11-22T00:40:58.288372
499999 /type/author ... 2022-11-22T00:53:46.568372

[500000 rows x 9 columns]
Extracted rows : 500000
Purified Total rows: 499911
Data loaded successfully into the database!
Dropped duplicate rows: 399917
Process finished with exit code 0
```

```
def extract_data(file_path):
    'works': json_data.get('works')
    })
    if len(data) >= 500000: # Limit to 400,000 rows (adjust as needed)
        break

Run edition EETL.py

C:\Users\Hailsha\PycharmProjects\pythonProject\venv\Scripts\python.exe "C:\Users\Hailsha\PycharmProjects\Data Mining\edition EETL.py"
type ... works
0 /type/edition ... [{"key": "/works/OL895114W"}]
1 /type/edition ... [{"key": "/works/OL1498779W"}]
2 /type/edition ... [{"key": "/works/OL14983278W"}]
3 /type/edition ... [{"key": "/works/OL14983191W"}]
4 /type/edition ... [{"key": "/works/OL28282135W"}]
...
499995 /type/edition ... [{"key": "/works/OL22412791W"}]
499996 /type/edition ... [{"key": "/works/OL9451097W"}]
499997 /type/edition ... [{"key": "/works/OL22413213W"}]
499998 /type/edition ... [{"key": "/works/OL22413464W"}]
499999 /type/edition ... [{"key": "/works/OL22413493W"}]

[500000 rows x 14 columns]
Extracted Rows: 500000
Data cleaned successfully!
No new data to load (all records were duplicates).
Process finished with exit code 0
```

```
def clean_data(data):
    # Drop missing or empty work_id or book_id
    data = data.dropna(subset=['work_id', 'book_id'])
    data = data[data['book_id'] != ''] # Discard rows with null or empty book_id

Run rating EETL.py

C:\Users\Hailsha\PycharmProjects\pythonProject\venv\Scripts\python.exe "C:\Users\Hailsha\PycharmProjects\Data Mining\rating EETL.py"
work_id book_id rating date
1 OL1629179W OL22581670H 5 2018-06-20
2 OL4226936W OL1068042H 5 2018-06-20
3 OL5264255W OL2719185H 5 2018-06-20
4 OL1681415W OL2582724H 5 2018-06-20
6 OL1853596W OL2331075H 2 2018-06-20
...
399995 OL28765887W OL10680590H 3 2023-01-18
399996 OL20882694W OL28419727H 5 2023-01-18
399997 OL4320394W OL2836669H 5 2023-01-18
399998 OL5885273W OL3472304H 4 2023-01-18
399999 OL5267298W OL25435142H 1 2023-01-18

[342856 rows x 4 columns]
Purified Data Rows: 342856
Data loaded successfully into the database!
Process finished with exit code 0
```

2: Data Loading and Analysis

1. Data Warehouse Implementation and Loading:

The screenshot displays two instances of the pgAdmin 4 interface, showing SQL queries and their results in a data warehouse.

Top Instance:

- Query:**

```
1 select * from dim_works ;
2 select * from dim_authors;
3 select *from dim_ratings;
4 select * from dim_editions;
5 select *from fact_book_ratings;
6
7
8 --relating authors works
9 SELECT *
10 FROM dim_authors
11 JOIN dim_works
12 ON dim_authors.author_id = dim_works.authors;
13
```
- Data Output:** A table with 10 rows and 8 columns. The columns are: type, author_id [PK], revision, created_at, name, key, and source_records. The data shows authors and their associated works.

Bottom Instance:

- Query:** The same query as the top instance.
- Data Output:** A table with 10 rows and 6 columns. The columns are: type, book_id [PK], revision, last_modified, and title. The data shows editions of books and their titles.

FileObjectToolsEditViewWindowHelp

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Query Query HistoryScratch Pad

```
122
123
124 select *from fact_book_ratings;
125
126 =====
127 =====
128
129
130
131
```

Data OutputMessagesNotifications

Showing rows: 1 to 259Page No: 1 of 1

	book_id character varying	title character varying (255)	work_id character varying	author_id character varying	author_name character varying (255)	rating numeric
1	OL11353750M	Sacramentary	OL7893135W	OL2589641A	Catholic	3
2	OL11466390M	What the Bible is All about	OL17048540W	OL20625A	Henrietta C. Mears	1
3	OL15233460M	De superman au surhomme	OL15653145W	OL20735A	Umberto Eco	3
4	OL15233460M	De superman au surhomme	OL15653145W	OL20735A	Umberto Eco	4
5	OL17286871M	Yanlis okumalar	OL15653145W	OL20735A	Umberto Eco	3
6	OL17286871M	Yanlis okumalar	OL15653145W	OL20735A	Umberto Eco	4
7	OL25065956M	Strongest and weakest	OL9592680W	OL1394228A	Camilla De la Bédoyère	4
8	OL25299695M	Democracy in the poetry of Walt Whitman	OL18208232W	OL7106853A	Thomas Riggs	5
9	OL27027057M	Contemporary theatre, film and television	OL18208232W	OL7106853A	Thomas Riggs	5
10	OL27564969M	Discover the Savage World	OL9592680W	OL1394228A	Camilla De la Bédoyère	4
11	OL35354818M	Beasties in my backyard	OL9592680W	OL1394228A	Camilla De la Bédoyère	4
12	OL35362368M	Baudolino	OL15653145W	OL20735A	Umberto Eco	3

Total rows: 259Query complete 00:00:00.182CRLF Ln 124, Col 1

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Query Query HistoryScratch Pad

```
122
123
124 select *from fact_book_ratings;
125
126 =====
127 =====
128
129
130
131
```

Data OutputMessagesNotifications

Showing rows: 1 to 259Page No: 1 of 1

	book_id character varying	title character varying (255)	work_id character varying	author_id character varying	author_name character varying (255)	rating numeric
1	OL11353750M	Sacramentary	OL7893135W	OL2589641A	Catholic	3
2	OL11466390M	What the Bible is All about	OL17048540W	OL20625A	Henrietta C. Mears	1
3	OL15233460M	De superman au surhomme	OL15653145W	OL20735A	Umberto Eco	3
4	OL15233460M	De superman au surhomme	OL15653145W	OL20735A	Umberto Eco	4
5	OL17286871M	Yanlis okumalar	OL15653145W	OL20735A	Umberto Eco	3
6	OL17286871M	Yanlis okumalar	OL15653145W	OL20735A	Umberto Eco	4
7	OL25065956M	Strongest and weakest	OL9592680W	OL1394228A	Camilla De la Bédoyère	4
8	OL25299695M	Democracy in the poetry of Walt Whitman	OL18208232W	OL7106853A	Thomas Riggs	5
9	OL27027057M	Contemporary theatre, film and television	OL18208232W	OL7106853A	Thomas Riggs	5
10	OL27564969M	Discover the Savage World	OL9592680W	OL1394228A	Camilla De la Bédoyère	4
11	OL35354818M	Beasties in my backyard	OL9592680W	OL1394228A	Camilla De la Bédoyère	4
12	OL35362368M	Baudolino	OL15653145W	OL20735A	Umberto Eco	3

Total rows: 259Query complete 00:00:00.182CRLF Ln 124, Col 1

File Object Tools Edit View Window Help

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Query Query History Scratch Pad x

```
1 select * from dim_works ;
2 select * from dim_authors;
3 select *from dim_ratings;
4 select * from dim_editions;
5 select *from fact_book_ratings;
6
7
8 --relating authors works
9 SELECT *
10 FROM dim_authors
11 JOIN dim_works
12 ON dim_authors.author_id = dim_works.authors;
13
```

Data Output Messages Notifications

Showing rows: 1 to 1000 Page No: 1 of 377

	type	work_id	revision	last_modified	title
	text	[PK] text	bigint	timestamp without time zone	text
1	work	OL10000115W	4	2019-03-10 02:42:04.184678	Litterature et reportage
2	work	OL10000144W	2	2020-12-05 02:23:24.753437	Safar fi wujuh
3	work	OL10000169W	3	2010-04-28 06:54:19.472104	Mathématiques, analyse, MP (cours, méthodes, exercices résolus)
4	work	OL10000206W	5	2011-11-25 06:14:28.313274	Grammaire critique de l'italien, volume 13
5	work	OL10000376W	3	2010-04-28 06:54:19.472104	Comme des garçons
6	work	OL10000435W	5	2021-02-24 05:15:17.154746	Ces grands discours qui ont fait le discours
7	work	OL10000494W	3	2010-04-28 06:54:19.472104	Lorie de A à Z
8	work	OL10000606W	7	2022-07-26 07:11:15.619585	L'Avant-scène Opéra, numéros 135-136
9	work	OL10000664W	3	2010-04-28 06:54:19.472104	La nutrition. médecine des suppléments alimentaires
10	work	OL10000706W	4	2022-09-14 15:47:04.700457	C'est la rhéa

Total rows: 376715 Query complete 00:00:02.120 CRLF Ln 1, Col 1

File Object Tools Edit View Window Help

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Second DB/postgres@Projects

Query Query History Scratch Pad x

```
1 select * from dim_works ;
2 select * from dim_authors;
3 select *from dim_ratings;
4 select * from dim_editions;
5 select *from fact_book_ratings;
6
7
8 --relating authors works
9 SELECT *
10 FROM dim_authors
11 JOIN dim_works
12 ON dim_authors.author_id = dim_works.authors;
13
```

Data Output Messages Notifications

Showing rows: 1 to 1000 Page No: 1 of 432

	type	book_id	revision	last_modified	title
	text	[PK] text	bigint	timestamp without time zone	text
1	edition	OL10000241M	3	2010-04-13 02:52:51.925831	The National Register of Apparel Manufacturers, Men & Boys
2	edition	OL10000488M	3	2011-04-28 06:17:32.348931	Public Processions Etc. (Northern Ireland) Bill [H.L.]
3	edition	OL10000723M	4	2022-07-17 04:38:18.718745	Crime and Disorder Bill [H.L.]
4	edition	OL10001042M	3	2011-04-28 19:27:40.205073	Northern Ireland Bill
5	edition	OL1000267M	11	2024-08-06 14:00:42.514369	The psalm killer
6	edition	OL10002693M	3	2011-06-08 04:28:03.025393	Health Authorities Bill
7	edition	OL10003167M	3	2011-04-28 00:52:09.260425	Nursery Education and Grant-Maintained Schools Bill
8	edition	OL10003490M	7	2022-07-17 05:19:12.575779	Arbitration Bill [Lords] (Parliamentary Debates: [1995-96])
9	edition	OL10003761M	4	2022-07-17 07:01:12.718767	Finance Bill (Except Clauses 2, 28 and 99) (Parliamentary Debates: [1998-99])
10	edition	OL10004076M	4	2022-07-17 07:01:03.810777	Welfare Reform and Domicile Bill (Parliamentary Debates: [1998-99])

Total rows: 431083 Query complete 00:00:02.693 CRLF Ln 4, Col 28

pgAdmin 4

Welcome perfect last.sql* x hhhhhh.sql* x Second DB/postgr... x Second DB/postgr... x Second DB/postgr... x Second DB/postgr... x Second DB/postgr... x Second DB/postgr...

Second DB/postgres@Projects

Query Query History

```
1 select * from dim_works ;
2 select * from dim_authors;
3 select *from dim_ratings;
4 select * from dim_editions;
5 select *from fact_book_ratings;
6
7
8 --relating authors works
9 SELECT *
10 FROM dim_authors
11 JOIN dim_works
12 ON dim_authors.author_id = dim_works.authors;
13
```

Data Output Messages Notifications

Showing rows: 1 to 1000 Page No: 1 of 500

	type	author_id	revision	created_at	name	key	source_records
	text	[PK] text	bigint	timestamp without time zone	text	text	text
1	/type/author	OL10000226A	1	2021-12-26 21:42:09.238563	Alan R. Blackmer	OL10000226A	(bwb:9780674593473)
2	/type/author	OL10000440A	1	2021-12-26 22:24:12.304187	Carmen Mellizo Sanz	OL10000440A	(bwb:9788490150221)
3	/type/author	OL10000917A	1	2021-12-26 23:32:38.430686	Tom Popkewitz	OL10000917A	(bwb:9781607527602)
4	/type/author	OL10001265A	1	2021-12-27 00:25:38.211432	Fan Chengda	OL10001265A	(bwb:9780295802060)
5	/type/author	OL10001270A	1	2021-12-27 00:25:45.312956	Mario R. Dinunzio	OL10001270A	(bwb:9781604264043)
6	/type/author	OL10001396A	1	2021-12-27 00:57:39.652969	Thomas S. O'Connor	OL10001396A	(bwb:9781606493557)
7	/type/author	OL10001709A	1	2021-12-27 01:38:04.408499	Jon Vriesacker	OL10001709A	(bwb:9781480371354)
8	/type/author	OL10001836A	1	2021-12-27 01:48:07.23888	Peter Lieuwen	OL10001836A	(bwb:9781476871578)
9	/type/author	OL10002328A	1	2021-12-27 02:48:53.242959	Yvonne Velasquez	OL10002328A	(bwb:9780578158846)
10	/type/author	OL10002401A	1	2021-12-27 03:15:47.849785	Blank	OL10002401A	(bwb:9780578477504761)

Total rows: 499911 Query complete 00:00:01.798

CRLF Ln 2, Col 1

pgAdmin 4

Welcome perfect last.sql* x hhhhhh.sql* x Second DB/postgr... x Second DB/postgr... x Second DB/postgr... x Second DB/postgr... x Second DB/postgr... x Second DB/postgr...

Second DB/postgres@Projects

Query Query History

```
1 select * from dim_works ;
2 select * from dim_authors;
3 select *from dim_ratings;
4 select * from dim_editions;
5 select *from fact_book_ratings;
6
7
8 --relating authors works
9 SELECT *
10 FROM dim_authors
11 JOIN dim_works
12 ON dim_authors.author_id = dim_works.authors;
13
```

Data Output Messages Notifications

Showing rows: 1 to 1000 Page No: 1 of 377

	type	work_id	revision	last_modified	title
	text	[PK] text	bigint	timestamp without time zone	text
1	work	OL10000115W	4	2019-03-10 02:42:04.184678	Litterature et reportage
2	work	OL10000144W	2	2020-12-05 02:23:24.753437	Safar fi wujuh
3	work	OL10000169W	3	2010-04-28 06:54:19.472104	Mathématiques, analyse, MP (cours, méthodes, exercices résolus)
4	work	OL10000206W	5	2011-11-25 06:14:28.313274	Grammaire critique de l'italien, volume 13
5	work	OL10000376W	3	2010-04-28 06:54:19.472104	Comme des garçons
6	work	OL10000435W	5	2021-02-24 05:15:17.154746	Ces grands discours qui ont fait le discours
7	work	OL10000494W	3	2010-04-28 06:54:19.472104	Lorie de A à Z
8	work	OL10000606W	7	2022-07-26 07:11:15.619585	L'avant-scène Opéra, numéros 135-136
9	work	OL10000664W	3	2010-04-28 06:54:19.472104	La nutrition. medecine des suppléments alimentaires
10	work	OL10000704W	4	2022-08-14 15:47:04.790457	C'est la zèle

Total rows: 376715 Query complete 00:00:02.120

CRLF Ln 1, Col 1